Paulina Lilian Ruíz Servín



Contact

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Profile

Mechatronics and Electrical Engineering student in the DHIK double-degree program (Tecnológico de Monterrey / Hochschule Zittau/Görlitz). Focus areas: electronics and firmware development (PCB design in Altium Designer, ESP32 firmware in C++, CAN/SPI/I2C/UART, analog measurement chains) and automation (PLC programming in TIA Portal, WinCC HMI, virtual commissioning in Process Simulate, robotics and machine vision). Team captain of a 30-member Shell Eco-marathon team; cleanroom experience at MIT.nano.

Education

09.2026 – present
Zittau, Germany

B.Eng. Electrical Engineering (DHIK double degree)

Hochschule Zittau/Görlitz

08.2023 – present
Mexico City

B.Sc. Mechatronics Engineering (English-taught program)

Tecnológico de Monterrey
Grade approx. 1.8 (92/100) • Merit scholarship (30 %)

Technical Experience

08.2023 – 08.2026
Mexico City

Team Captain, Electronics & Telemetry Lead, Mentor

Escudería EcoVolt CCM (Shell Eco-marathon)
Electronics & Design Member (2023–2024) → Electronics Co-Lead (2024–2025) → Team Captain and Telemetry Lead (2025–2026) → Mentor (2026)

- **Telemetry hardware:** co-founded and led the telemetry area (concept, first schematic, breadboard validation); design in Altium Designer (schematic, layout, BOM, bring-up): telemetry PCB Brazil 2025, lead developer of the multilayer board USA 2026 and INA240 current-sensing module.
- Bidirectional shunt current sensor from a home-etched prototype to a manufactured board.
- **Firmware (ESP32, C++):** CAN interface to the team's in-house motor controller (incl. transceiver circuit), SPI, I2C, UART and SD-card drivers, current and voltage acquisition; GPS configuration (NEO-6M, u-center); defined the measured signals together with the data team.
- **HIL test bench (team work):** Python bench with a driving simulator (Assetto Corsa), telemetry and a real traction motor; result: constant throttle more efficient than pulse-and-glide.
- SMD/THT soldering, characterization of transistors and diodes (Zener, Schottky, rectifier), lab instrumentation (oscilloscope, function generator, bench power supply, multimeter); planning and wiring of the vehicle incl. emergency stop, support with battery assembly.
- **Ergonomics:** analysis of the driver's position with a virtual human model in Tecnomatix Process Simulate (RULA assessment) to validate cockpit and seating position.
- **Tool tracking:** concept and Python/OpenCV pilot of a QR-based inventory system (QR decoding, ID validation, logging) for paddock logistics.
- Led the 30-member team as team captain (2025–2026); driver of the prototype in competition and test runs.
- **Team achievements:** 1st place (2025) and 2nd place (2026) Data & Telemetry Award; Vehicle Design Award (2026); Honorable Mention, Altair Global Student Contest (2025).

10.2025 – 10.2025
Cambridge, USA

Visiting Student – Intensive Program (Semiconductor Fabrication)

MIT.nano Lab, Massachusetts Institute of Technology

- Silicon solar cell (Si_3N_4 anti-reflection coating, photolithography, RIE, Ti/Al sputtering) with 15.87 % efficiency from my own IV measurement.
- MEMS cantilevers and microfluidic mixers (silicon, PDMS) in the cleanroom (class 100/1K/10K): thin-film deposition (PECVD/PVD), wet-chemical and plasma etching (RCA, BOE, HF) and process documentation.

Projects

02.2026 – 06.2026
Mexico City

Digital Twin & PLC Programming of a CNC Manufacturing Cell

ORZA Tech / Tecnológico de Monterrey (team project, 3 people; own station)

- PLC program in TIA Portal (ladder) for a virtual S7-1500 (CPU 1511-1 PN) in PLCSIM Advanced: 35-step sequence, interlocks, emergency stop, manual/automatic mode, recipes (FIFO) and quality routing (good part, rework, scrap); KPIs and cycle log in a retentive DB. **Result:** complete cycles commissioned fully virtually.
- Digital twin in Process Simulate (3× UR5, AGV, conveyors, CNC mill), all sensors/actuators mapped to PLC I/O; UR5 program (PolyScope, URSim VRC) synchronized via I/O handshake.
- WinCC HMI and cell-wide SCADA (OEE, cycle time, scrap rate); S7 GET/PUT with connection alarm, Modbus TCP with my own C# sensor simulator.
- CAM program in SOLIDWORKS CAM (6 tools, cycle approx. 12.8 min) with G-code analysis, plus a turning operation; Cognex In-Sight inspection validated against a caliper (±3 %); pick-and-place exercises on a physical UR5e.

02.2026 – 06.2026
Mexico City

Electronics & Machine Vision for a SCARA Robot (4 DoF)

Tecnológico de Monterrey (team project, 4 people)

- **Result:** 5 of 6 parts sorted autonomously, approx. 14 s per pick-and-place cycle.
- Control board in Altium Designer (single-layer): ESP32, stepper and DC drivers, encoders; selection of motors and drivers; integration and validation of the homing sensors (limit switches, AS5600 encoders) against known reference angles.
- Machine vision in Python (OpenCV, YOLOv11n with mAP@0.5 = 0.72) and ArUco homography (pixels to mm); grasp coordinates sent via MQTT to the ESP32 firmware.
- Own forward/inverse kinematics in MATLAB; PID joint control in MATLAB/Simulink as model-in-the-loop.

08.2025 – 12.2025
Mexico City

Automated Viscosity Measurement System with Sorting

Endress+Hauser / Tecnológico de Monterrey (team project, 4 people)

- Designed all three boards of the measurement station in Altium Designer: ESP32 master and slave (two-layer, IPC-2221, manufactured by JLCPCB) and the amplifier board.
- Two-stage op-amp measurement chain (1 Ω shunt, gain approx. 400) mapping a 6 mA window of the motor current onto the ESP32 ADC; simulated in Multisim, verified on the board. **Result:** reliable separation of 252 cP and 1676 cP.
- Co-developed the ESP32 firmware (C++): homing, measurement routine with moving average, I2C color sensor, UART between master and slave, Bluetooth to the LabVIEW HMI.
- Co-programmed the conveyor sequence in ladder logic (OpenPLC); helped build the conveyors (5 and 8 kg payload vs. $<$ 1 kg required), placed and calibrated the IR sensors.

02.2025 – 06.2025
Mexico City

Scissor Lift Table with Automated Guided Vehicle (AGV)

Tecnológico de Monterrey (team project, 5 people)

- Load-cell amplifier board without an off-the-shelf amplifier IC (Wheatstone bridge, differential and non-inverting amplifiers, RC low-pass): linear 0–5 kg to 0–3 V at the ESP32 ADC.
- Concept for splitting the control across three ESP32; 16×2 LCD driver in C++ (ESP-IDF); commissioning of the lift drive (NEMA 17, DRV8825).
- Full assembly in SolidWorks with motion study of the tilting mechanism and static analysis (max. deflection 0.28 mm); motion simulation in Siemens NX; cam for $\geq$ 30° tilt angle ($\geq$ 25° required) designed and 3D-printed.
- Machined the motor coupling and connecting rod (aluminum) on lathe and mill; base frame loaded with over 100 kg, 7 kg in the final demo (5 kg required).

Publications

09.2026

Real-Time Telemetry and Computational Intelligence Pipeline for Energy-Efficient Driving of an Electric Eco-Marathon Vehicle

SAIBOT – IEEE Latin American School on Computational Intelligence and Robotics
P. L. Ruíz-Servín, J.D. González-Fernández, A. Flores-Mendoza, J. Izquierdo-Reyes • *Submitted, under review*

Extracurricular Activities

04.2025 – 05.2026	**Vice President of Finance**, SAIMT (mechatronics student society): event organization, quotations, sponsor meetings, budget and bill of materials (BOM).
06.2024 – 06.2026	**PEER Mentor**, Tecnológico de Monterrey: mentoring first-year students to strengthen problem-solving and communication skills.
02.2026	**Panelist**, International Day of Women and Girls in Science (Ingenia).

Skills

Languages	Spanish (native), English (C1), French (B1), German (B1)
Electronics	Altium Designer, PCB design, op-amp signal conditioning, current sensing, Multisim, soldering (SMD/THT)
Test & Measurement	Bring-up, failure analysis, oscilloscope, function generator, multimeter, technical documentation
Embedded	ESP32, C/C++, FreeRTOS, ESP-IDF, PlatformIO
Communication	CAN, SPI, I2C, UART, MQTT, Bluetooth, Modbus TCP, S7 GET/PUT
Control & Simulation	PID and nonlinear control, cascade control, MiL/HiL; MATLAB/Simulink/Simscape
Automation	TIA Portal, Siemens S7-1500 and LOGO!, WinCC HMI, PLCSIM Advanced, Studio 5000 (ControlLogix), FactoryTalk Optix, OpenPLC; ladder logic
Digital Twin	Tecnomatix Process Simulate, virtual commissioning, ergonomics (RULA)
Robotics & Vision	Universal Robots (UR5/UR5e, PolyScope, URSim), Cognex In-Sight, OpenCV, YOLO, ArUco
CAD & Manufacturing	SolidWorks CAD & CAM, Siemens NX, Fusion CAM, G-code; turning, milling, 3D printing
Programming	C/C++, Python, C#, SQL, MATLAB, LabVIEW, LaTeX

Certificates

2026	• Altium: Educational PCB Basic Design • MathWorks: MATLAB/Simulink/Simscape Onramp • Universal Robots: Educational Robotics – Core • Ansys: Intro to Ansys Maxwell & HFSS • Siemens: Process Simulate Associate • Dassault Systèmes: SolidWorks CSWA • Rockwell/Cisco: Industrial IoT
2025	• MIT.nano: Certificate (cleanroom semiconductor fabrication)
Languages	• Goethe-Zertifikat A2 (German) • DELF B1 (French)

Interests

Motorsport (MotoGP, Formula 1) • Aviation • Intercultural exchange • Travel